

Summary

The paper asks an interesting, forward-looking question, namely whether the recent surge in inflation is associated with an increase in stagflation risk. The author identifies the nominal stock-bond return correlation as a promising indicator as to whether economic downturns are associated with higher inflation or not. As documented in the author's earlier work, the sign of this correlation flips from positive to negative around 2000. Building on that observation, the current paper asks which structural changes can induce this change in the stock-bond return correlation pattern and then evaluates whether it is likely that these structural changes have reversed in the recent episode of higher inflation.

To identify these structural changes, the paper manages to combine a relatively standard three equation New Keynesian model with a non-linear asset pricing framework based on habit preferences. Despite its simplicity, this model is able to speak to a rich set of macro and asset pricing moments. The model is evaluated using two different parameterizations, one for the pre-2000 period with a positive stock-bond beta and for the post-2000 period where the opposite is true. There are two key differences across parameterizations: First, in the pre-2000 period, supply shocks are more prevalent, while demand shocks dominate the second sample period. Second, monetary policy is more inflation focused and features less inertia in the earlier period. In counterfactual analysis, the author finds that a reversal of both of these parameter changes would be necessary in order to again observe a positive stock-bond beta. The conclusion is that the recent emergence of supply shocks is unlikely to return the economy to the 1980s, as long as policy is not also changing again.

Main comments

I very much enjoyed reading this paper and appreciate the ambition to use asset price moments to differentiate between macroeconomic models. I have three related concerns: First, the marginal contribution relative to the author's earlier work (Campbell, Pflueger and Viceira 2020) seems limited. Second, the quantitative success of the model vis-a-vis the data is a bit unclear or unconvincing. This seems important when evaluating the counterfactual results. Third, the paper could provide more economic analysis and explanation as to why the stock-bond beta is the right indicator to measure perceived stagflation risk. My overall suggestion is to strengthen the economic analysis as well as the quantitative fit, which would then also elevate the contribution of the paper.

Contribution relative to earlier work

The paper builds on Campbell, Pflueger and Viceira (2020). This earlier paper highlights the time variation in the stock-bond return correlation and provides a model to rationalize this observation. In that model, inflation and the output gap are exogenous processes, and the correlation of those two processes governs the stock-bond beta. Here instead, the stochastic processes are governed by a structural model but mathematically and conceptually the difference seems relatively small. Indeed, the previous paper, contains language such "Keeping in mind that (21) represents equilibrium dynamics and not a structural model, the increase in b_{ix} from negative to positive is suggestive of an increase in the Fed's interest in stabilizing output in period 2, relative to the strong focus on stabilizing inflation in period 1." The conclusion of that paper provides an outline of how to use the stock-bond beta in order to distinguish macro models and how to setup a small-scale NK model with habit preferences.

If someone else than one of the authors of that earlier paper had written this present paper, it would rightfully struggle to receive recognition as the earlier paper provides the key insights and ideas present in this work. Here, this is less clear, given that one of the original authors is implementing the discussed extension, but I still think that the new paper has to be evaluated given its marginal contribution. While it is a very nice paper in itself, I find the distance between the papers to be quite small.

Quantitative fit and calibration strategy

The paper's calibration strategy is to pick a set of macro and asset pricing moments and choose parameters to fit these moments as best as possible. There are more moments than parameters so that the model is overidentified and the model's quantitative success is evaluated through its ability to jointly match all moments. Unfortunately there is no single table which compares the twelve targeted moments in the model with their data counterparts, but I was surprised to see that the model fit for most key moments is not particularly good. For example, the standard deviation of consumption growth is more than 30% too low in the first period, and about as much too high in the second period, while in the data consumption growth volatility remains stable. The nominal rate is more volatile in the data, in particular more than twice as high in the second period, and the model undershoots the slope-excess return predictability regression coefficients substantially, with even the wrong sign in the second period.

As I discuss below, there is an existing literature that tries to jointly match term structure and macro models. I am not sure how the quantitative success of this model compares to earlier work, especially since different moments are targeted. However, for this particular exercise, quantitative success seems key in order to then be able to use the model for counterfactual analysis. Since the structural model is the key contribution relative to the earlier work, success in this dimension seems even more important.

I assume that the calibration algorithm struggles to get the lead-lag coefficients right, which have relatively narrow standard errors but for whose fit the model does not have sufficient dynamic complexity. Struggling to fit these moments, the algorithm then sacrifices the fit of moments that would actually be easy to target such as consumption and nominal rate volatility. While I am very sympathetic to target a larger set of moments than parameters, I would eliminate the lead-lag relationships for which the model might not be well suited and instead focus on more standard volatilities and correlations of output, inflation and asset pricing moments. If the stock-bond return correlation is indeed a key identifying moment, it seems reasonable to also include it as well as a target (I would think that the advantage of an overidentified calibration is that success is not measured through untargeted moments but through the joint match of a larger set of moments). Given those moments one could then present a table with all targets and their fit and discuss clearly the extent to which the model is successful and should therefore be considered a useful laboratory for counterfactual analysis.

The stock-bond beta

As the paper states at the beginning of the counterfactual analysis, "the beta of nominal bonds is of particular interest as an indicator of the risks of high inflation recessions". While this is the key idea of the paper, I did not fully understand this premise. To clarify this point, consider the model environment with monetary policy as calibrated for the first sample period. The paper

discusses that in this environment the stock-bond beta is positive when supply shocks dominate but negative when demand shocks are more prevalent. However when doing this exercise, the author changes both the agents' expectations over the prevalence of shocks as well as realizations of shocks themselves.

Consider instead an exercise where agents in the 1980s are very concerned that a rare but severe supply shock could happen, as they might remember from the 1970s. However, in the simulation only demand shocks are occurring. My guess is that in that simulated sample, one would measure a negative stock-bond beta, driven by the demand shocks, even as investors are actually concerned about high inflation recessions. This thought experiment, I think, shows that the stock-bond beta is not a particularly forward looking moment.

The ideal moment would of course be a measure of the expected term premium, which forward looking quantifies to what extent nominal bonds are perceived to be risky. I understand the authors argument that is hard to measure the expected term premium in a finite sample period and I agree that it is worthwhile to entertain the idea that the expected term premium has been negative since 2000 (even if portfolio managers at SVB seem not to have come to that conclusion). The paper, however, did not clarify sufficiently to me in what way the stock-bond correlation has a forward looking component to it. The latter seems crucial if one wants to take currently positive stock bond betas as an indicator for whether the recent surge in inflation has increased the risk of a stagflation.

Let me clarify that I do appreciate that there are several interesting discussions along those line in the paper. It is, for example, striking that in the post 2000 calibration a switch back to higher supply shocks induces an even more negative nominal stock-bond beta (which seems a bit contradicted by recent data, where the stock-bond beta has been increasing and news about receding inflation have been associated stock market gains). Figure 8 sheds some light on the importance of the policy rule in driving the nominal stock-bond beta, but the economic analysis remains a bit limited and similar discussions are already present in Rudebusch and Swanson (2012).

I have not much suggestions on how to deepen the economic insights here, except for maybe clarifying to what extent there is a forward looking element in stock-bond beta by holding fixed agents expectations while shock prevalences change or vice versa. This would allow a discussion of how perceived risks rather than realized shocks drive this statistic.

Additional comments

- The conclusions of the paper load largely on the role of monetary policy. This importance on an exogenously varying policy rule seems somewhat frustrating, since the high inflation responsiveness of policy in the 1980s was presumably a response the economic outcomes in the 1970s, which required a policy rule to bring down sticky inflation expectations. Part of the severeness of a stagflation episode may come from the fact that when the inflation expectation process changes, the policy rule needs to adapt.
- The fragility of the positive stock-bond beta with respect to the policy rule seems to mirror the struggles of earlier general equilibrium macro models to feature an upward sloping yield curve. I would argue that the paper could connect more clearly to these papers (for example Rudebusch and Swanson (2012, AEJ:Macro), van Binsbergen et al. (JME, 2012), Kung (JFE, 2015)). All of these papers jointly match macro and term structure moments and they could be used in a similar way that the model presented here is used. Maybe a comparison to these papers could also help to clarify the quantitative success of the present paper.

- I could not find a description of how the model moments are calculated. I believe the right way to do so is to simulate several short samples that have the same length as the considered data periods. Most importantly, I think one should start the 1980s sample from very high inflation expectations as was probably the case in the data, which will presumably affect simulated moments given the adaptive expectations.
- Given that stagflation was an important phenomenon of the 1970s, in the presence of a very different policy rule, it might be worthwhile to also add the period of 1960-1980?
- Just as I do not understand the stock-bond beta as a forward looking stagflation risk measure, it is also not clear to me how the following statement on the first page can be model-independent true: "During the 1980s nominal bond betas were positive nominal bonds were risky whereas during the 2000s nominal bond betas were negative nominal bonds were safe." First, I would argue betas are a measure of realized shocks, as discussed above, but I think such a statement seems to also rely too much on the stock market being highly correlated with the actual stochastic discount factor.